



MetaboCensoR Tutorial

2026-09-04

Tutorial Edition 1.5

Software Release 1.5.x



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General Information

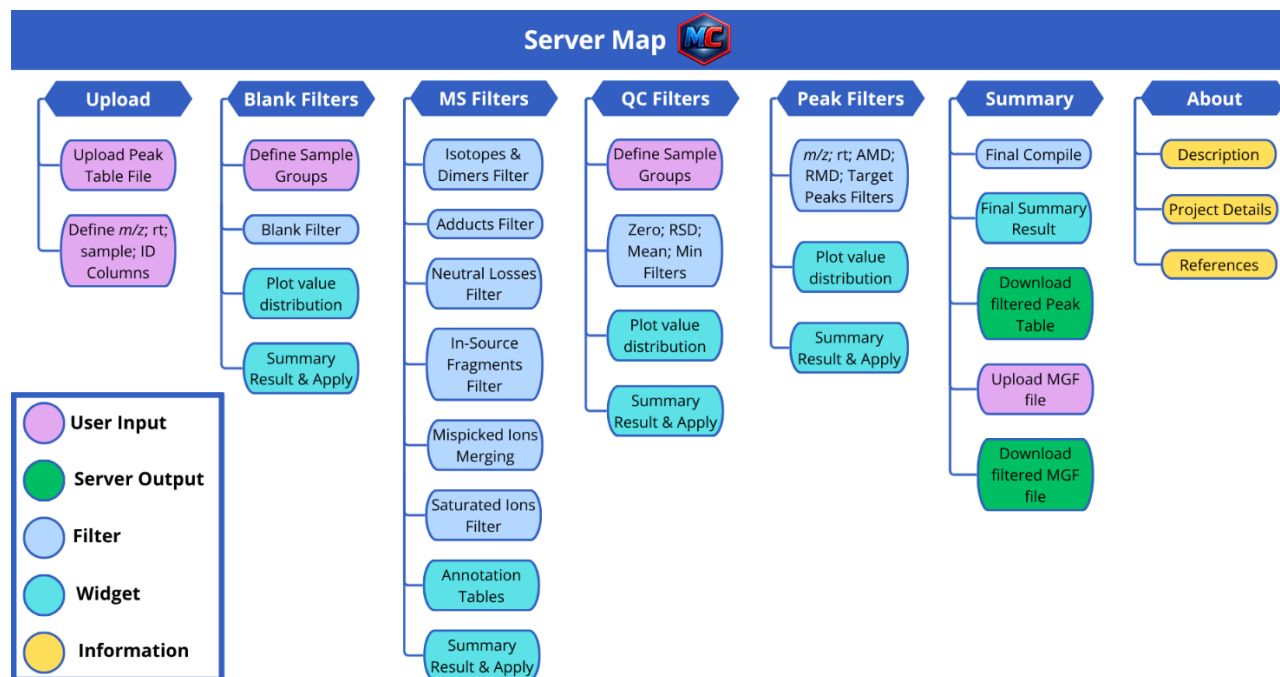
This tutorial aims at guiding the user to process data LC-MS data through Shiny GUI interface of MetaboCensoR. MetaboCensoR is an input-versatile Shiny application and local R package for analyte-centric peak table filtering. The workflow integrates four complementary modules for blank filtering, redundant ion-species filtering, quality-control filtering, and peak-based filtering. MetaboCensoR also provides interactive threshold optimization, exportable annotation tables, and synchronized filtering of associated .mgf files.

An overview of MetaboCensoR is presented below. The workflow is organized into six functional tabs that collectively support peak table curation, parameter optimization, and export of filtered results. The output of the app is a filtered peak table and, optionally, an MGF spectra file. Only the peak table upload is required for full workflow execution and final output generation. All other tabs are optional.

License: GPL-3.0

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Project Repository: github.com/plyush1993/MetaboCensoR





Running & Installation

Shiny Web Deployment is available by the link:

plyush1993.shinyapps.io/metabocensor/

Local installation of R package:

```
if (!require("BiocManager", quietly = TRUE)) {  
  install.packages("BiocManager")  
}  
if (!require("remotes", quietly = TRUE)) {  
  install.packages("remotes")  
}  
remotes::install_github("plyush1993/MetaboCensor", INSTALL_opts = "--no-multiarch")
```

or

```
if (!requireNamespace("pak", quietly = TRUE)) install.packages("pak")  
pak::pak("plyush1993/MetaboCensor")
```

Run Locally:

```
MetaboCensor::run_metabocensor()
```

Input Requirements

The application accepts peak tables in **.csv** format and MS1/MS2 mass spectral data in **.mgf** format, both of which can be directly exported from dedicated metabolomics processing software (xcms, MZmine, MS-DIAL). Make sure that your input peak table is aligned with data formats. Examples of acceptable formats are available here: [Input Examples](#).

Important note

The Shiny server has limited memory capacity (**1 GB**). Computationally intensive steps (like MS filters) or large MGF files may cause the app to stack or disconnect. For large datasets or memory-intensive analyses, we recommend activating the “**Limit table previews**” option and running *MetaboCensor* locally



Overview of Interface

The screenshot displays the MC interface with three main panels highlighted by red boxes and numbers:

- 1. Sidebar Panel (Control Panel):** Contains an 'Upload' section with a 'Data table type' dropdown (set to 'mzMine') and an 'Upload Peak Table (*.csv)' section with a 'Browse...' button. Below this is a 'Global parsing settings' section with dropdowns for 'Feature ID (auto-generated)', 'm/z column', and 'rt column'. At the bottom, there are radio buttons for 'How to define sample columns?': 'Auto-detect numeric sample columns', 'By keyword match' (selected), and 'Pick columns manually'.
- 2. Main Panel (Display Area):** Displays a dataset table titled 'Dataset: Example'. The table has columns: row ID, row m/z, row retention time, row ion mobility, row ion mobility unit, row CCS, correlation group ID, annotation network number, best ion, auto MS2 verify, identified by n, partners, and neutral M mass. The table shows 6 entries. Above the table, it says 'TOTAL SAMPLES DETECTED: 9' and 'TOTAL FEATURES (PEAKS): 6,167'. Below the table, it says 'Showing 1 to 6 of 6,167 entries' and has pagination controls.
- 3. Navigation Bar (Tab Menu):** Located at the top, it contains tabs: 'Upload Data', 'Blank Filters', 'MS Filters', 'QC Filters', 'Peak Filters', 'Final Summary', and 'About'.

GUI is organized into 3 main panels:

1. The Sidebar Panel (Control Panel)

- This is the primary area for user interaction. It houses the input controls where users upload data, configure global settings, define parameters, and execute filtering steps.

2. The Main Panel (Display Area)

- This is where the application renders the output and displays interactive data tables, statistical plots, and provides summary metrics.

3. The Navigation Bar (Tab Menu)

- This allows users to navigate linearly through the sequential steps of the data processing pipeline, from data upload to final summary and export.

Contextual assistance is embedded throughout the interface. Users can access detailed parameter descriptions and recommendations by interacting with the tooltip icons.

On the following pages, yellow numbered circles  indicate the recommended operational order within each interface tab in *Control Panel*.

Red numbered squares  describe widgets in *Display Area*.



Implementation Details

All filter groups are applied sequentially in the order shown in the tab panel: Blank Filters → MS Filters → QC Filters → Peak Filters. Changes made on later tabs do not affect the results of previous tabs. To change the order of applied filters (e.g., 'Peak Filters' before 'Blank Filters'), run the application sequentially several times, using the output peak table from one run as the input for the next.

Quick Start

Press button  in “Upload data” tab to upload attached example dataset and get access to the MetaboCensoR functionality. If you use this example, run the Blank filter first to save memory for further steps. You can keep all parameters by default.

Briefly, it is the LC-MS plant profiling dataset, which is described in [Houriet et al.](#) Briefly, methanol extracts from the plant ashwagandha [*Withania somnifera* (L.) Dunal] together with blanks were analyzed on RP column in DDA positive mode on Thermo Q-Exactive Plus Orbitrap. Raw mzML files were then processed in mzMine in default pre-settings for UPLC-DDA to obtain resulted peak table.



Tab “Upload Data”: Control Panel

The screenshot shows the 'Upload Data' control panel of the MetaboCensoR application. The panel includes a navigation bar with tabs: 'Upload Data', 'Blank Filters', 'MS Filters', 'QC Filters', 'Peak Filters', 'Final Summary', and 'About'. The 'Upload Data' tab is active. The panel is divided into two main sections: 'Upload' and 'Global parsing settings'.

Upload Section:

- 1. Pick the right table type and upload:** A dropdown menu for 'Data table type' is set to 'mzMine'. Below it is a button 'Browse...' and the text 'No file selected'.

Global parsing settings Section:

- 2. Specify peak characteristic columns:** Three dropdown menus are shown: 'Feature ID' (set to 'row ID'), 'm/z column:' (set to 'row m/z'), and 'rt column:' (set to 'row retention time').
- 3. Specify sample intensity columns:** A section titled 'How to define sample columns?' with three radio buttons: 'Auto-detect numeric sample columns', 'By keyword match' (selected), and 'Pick columns manually'.
- 4. Limit table previews:** A toggle switch labeled 'Limit table previews' is currently turned off.
- 5. Clear dataset:** An orange button labeled 'Clear dataset'.
- 6. Example dataset shortcut:** A blue button labeled 'Example dataset' with an upward arrow icon.

Summary Statistics:

- TOTAL SAMPLES DETECTED: 9
- TOTAL FEATURES (PEAKS): 6,167
- LARGE DATASETS / MEMORY LIMITS: (indicated by a warning icon)



Tab “Upload Data”: Display Area



[Upload Data](#)[Blank Filters](#)[MS Filters](#)[QC Filters](#)[Peak Filters](#)[Final Summary](#)[About](#)

Upload

Data table type: ?

mzMine

Upload Peak Table (*.csv)

Browse...

 No file selected

Global parsing settings

Feature ID:

row ID

m/z column:

row m/z

rt column:

row retention time

How to define sample columns?

☐ Auto-detect numeric sample columns

☒ By keyword match

☐ Pick columns manually

Sample column keywords (pick/add multiple):

.mzML .mzXML

Limit table previews ☐

Clear dataset

Example dataset

?

TOTAL SAMPLES
9

1 Statistics of uploaded dataset

Dataset: Example

Show 6 entries

Search:

	row ID	row m/z	row retention time	row ion mobility	row ion mobility unit	row CCS	correlation group ID	annotation network number
1	108	148.0277472	0.3691534				31	0
2	181	169.0462455	0.3698888				31	
3	175							20
4	231	216.9512638	0.371403				31	
5	43	234.961351	0.3725722				31	14
6	145	244.9475245	0.37269926				31	

Showing 1 to 6 of 6,167 entries Previous 1 2 3 4 5 ... 1,028 Next



Tab “Blank Filters”: Control Panel

Sample Label

Label source: 1

- ☒ From sample names (by token)
- ☐ From metadata CSV
- ☐ From custom CSV (one column, no header)
- ☐ Manual editable table

Token index: 2

Token separator: -

☒ Show labels table

Blank / Media Removal

Enable Blank filter: ☒

Choose blank/media group(s): Blank

Filtering mode:

- ☒ Blank < Cutoff * Sample Max
- ☐ Drop any feature detected in Blank

Cutoff (fraction of max experimental mean): 0.1

Plot values distribution

☒ Show Blank plot panel

Apply

Clear

Input table

Show 5 entries

row	ID	m
1	108	148.02774
2	181	169.04624
3	175	128.01976
4	231	216.95126
5	43	234.96133
6	145	244.94752

Output table — a

Show 5 entries

row	ID	m
1	108	148.02774
2	181	169.04624
3	175	128.01976
4	231	216.95126
5	43	234.96133
6	145	244.94752

Showing 1 to 5 of 2,597 entries

Previous 1 2 3 4 5 ... 520 Next

Filtering results — Step 1 (Blank Filters)

Metric	Value
Features before	6167
Removed by Blank Filters	3570
Features after	2597

1. Sample Labels — how to specify groups

Choose a label source:

- From sample name:

Define token and index for extracting labels from sample names.

Example — parse from sample names

Sample name: Sample_WT_1_R-C1_1_5037.mzML Peak area

Separator: _

Token index: 2

Resulting label: WT

- From metadata CSV:

Upload a metadata table with column names, choose the sample-name column and the column to use as Label. Rows are matched by sample name. You can also clean sample name to remove file extension, suffixes, etc.

- From custom CSV:

Upload a one-column CSV without header. Label order must match sample column order.

- Manual editable table:

The editable label table is filled from the current token-based labels. Double-click cells in the Label column to manually correct group names.

2. Show resulted Labels in Display Area

3. Set Up Blank Filters

Enable filter.

Specify blank groups.

Choose between filtering modes:

- Blank < Cutoff * Sample Max

- Drop any feature detected in Blank

4. Plot distribution of filtered peaks

5. Apply or Discard Filtering Step



Tab “Blank Filters”: Display Area

MetaboCampor

Upload Data

Blank Filters

MS Filters

QC Filters

Peak Filters

Final Summary

About

Sample Label

Label source:

From sample names (by token)

From metadata CSV

From custom CSV (one column, no header)

Manual editable table

Token index:

2

Token separator:

-

☒ Show labels table

Blank / Media Removal

Enable Blank filter

Choose blank/media group(s):

Blank

Filtering mode:

Blank < Cutoff * Sample Max

Drop any feature detected in Blank

Cutoff (fraction of max experimental mean):

0.1

Plot values distribution

☒ Show Blank plot panel

Apply

Clear

Input table

Show

5

 entries

Search:

	row ID	mz	rt	row ion mobility	row ion mobility unit	row CCS	correlation group ID	annotat network number
1	108	148.0277472	0.3691534				31	0
2	181	169.0462455	0.3699988				31	
3	175	128.0197664	0.37076914				31	20
4	231	216.9512638					31	
5	43	234.961351	0.3725				31	14

Showing 1 to 5 of 6,167 entries

Previous

1

2

3

4

5

...

1,234

Next

Check matching Sample Names and Group Labels:

Show

6

 entries

Sample	Label
Orbi_Blank_A.mzML Peak area	Blank
Orbi_Blank_B.mzML Peak area	Blank
Orbi_Blank_C.mzML Peak area	Blank
Orbi_Sample_A.mzML Peak area	Sample
Orbi_Sample_B.mzML Peak area	Sample
Orbi_Sample_C.mzML Peak area	Sample

Showing 1 to 6 of 9 entries

Previous

1

2

Next

Check Blank Ratio Distribution:

Blank to Sample Ratio Distribution

Count

Ratio: Mean(Blank) / Max(Mean(Sample)) (log10 scale)

Filtered: 57.9%

Output table — after Blank Filters

Show

5

 entries

Search:

	row ID	mz	rt	row ion mobility	row ion mobility unit	row CCS	correlation group ID	annotat network number
1	108	148.0277472	0.3691534				31	0
2	181	169.0462455	0.3699988				31	
4	231	216.9512638	0.371403				31	
5	43	234.961351	0.3725				31	14
6	145	244.9475245	0.37269				31	

Showing 1 to 5 of 2,597 entries

Previous

1

2

3

4

5

...

520

Next

Filtering results — Step 1 (Blank Filters)

Metric	Value
Features before	6167
Removed by Blank Filters	3570
Features after	2597

10



Tab “MS Filters”: Control Panel

The screenshot shows the 'MS Filters' tab in a software interface. The left sidebar contains several filter sections, each with a title and a set of controls. A yellow bracket on the left groups sections A through F. Numbered callouts (1-8) point to specific controls:

- 1. Enable & Set Up Isotopes Filtering**: Points to the 'A) Isotopes & Dimers' section. Controls include 'Enable isotope/dimer collapse' (toggle), 'Correlation $\geq$ ' (0.8), 'rt tolerance (min)' (0.01), 'm/z tolerance type' (Da selected), 'm/z tolerance (Da)' (0.005), 'Max charge (z_max)' (3), 'Max isotope order (n)' (1), and 'Dimer-series max n_d' (3).
- 2. Enable & Set Up Adducts Filtering**: Points to the 'B) Adduct Families' section. Controls include 'Enable adduct-family collapse' (toggle), 'Polarity' (positive selected), 'Adducts CSV (optional)' (Browse... button), 'Correlation $\geq$ ' (0.8), 'rt tolerance (min)' (0.005), 'm/z tolerance type' (Da selected), 'm/z tolerance (Da)' (0.005), and 'Minimum neutral mass' (50).
- 3. Enable & Set Up Neutral Losses Filtering**: Points to the 'C) Neutral Losses' section. Control is 'Enable neutral loss collapse' (toggle).
- 4. Enable & Set Up In-Source Fragments Filtering**: Points to the 'D) In-Source Fragments' section. Control is 'Enable in-source fragment collapse' (toggle).
- 5. Enable & Set Up Mispicked Ions Filtering**: Points to the 'E) Mispicked Ions' section. Control is 'Enable mispicking ions collapse' (toggle).
- 6. Enable & Set Up Saturated Ions Filtering**: Points to the 'F) Saturated Ions' section. Control is 'Enable ringing artifact collapse' (toggle).
- 7. Print & Download annotation tables**: Points to the 'Show isotope/dimer groups' checkbox.
- 8. Apply or Discard Filtering Step**: Points to the 'Apply' and 'Clear' buttons at the bottom.

The main area displays two tables: 'Input table' and 'Output table - after MS'. The 'Input table' shows 5 entries with columns: row ID, mz, rt, row ion mobility, row ion mobility unit, row CCS, correlation group ID, and annotation number. The 'Output table' shows 5 entries with columns: row ID, mz, and correlation group ID. A summary table at the bottom shows counts for various categories: Total (1909), Features (4258), and others.

1. Enable & Set Up Isotopes Filtering

Need parameter help? Click the tooltips or read the About tab for details. **Pro-Tip:** Start with the default settings.

2. Enable & Set Up Adducts Filtering

Need parameter help? Click the tooltips or read the About tab for details. **Pro-Tip:** Start with the default settings and check your polarity. You can upload custom adduct list in csv.

3. Enable & Set Up Neutral Losses Filtering

See next page.

4. Enable & Set Up In-Source Fragments Filtering

See next page.

5. Enable & Set Up Mispicked Ions Filtering

See next page.

6. Enable & Set Up Saturated Ions Filtering

See next page.

7. Print & Download annotation tables

8. Apply or Discard Filtering Step



C) Neutral Losses

Enable neutral loss filter ☒ ⓘ

Neutral loss CSV (optional)

Browse... Built-in list

Polarity

☒ positive ☐ negative

Correlation $\geq$

0.95

rt tolerance (min)

0.002

m/z tolerance type:

☒ Da ☐ ppm

m/z tolerance (Da)

0.005

D) In-Source Fragments

Enable in-source fragment filter ☒ ⓘ

Correlation $\geq$

0.95

rt tolerance (min)

0.002

☒ Control intensity ratio

frag/prec $\geq$

0.001

frag/prec $\leq$

2.5

☒ Require MS2 fragment match ⓘ

MGF file

Browse... orbi_iimn_gnps.mgf

Upload complete

MGF loaded: orbi_iimn_gnps.mgf (1965 spectra)

MGF Feature ID field:

acquisitionNum

MS2 relative intensity $\geq$ (% base peak)

1

MS2 fragment m/z tolerance (Da)

0.01

E) Mispicked Ions

Enable mispicking ions merging ☒ ⓘ

Correlation $\geq$

0.8

rt tolerance (min)

0.002

m/z tolerance type:

☐ Da ☒ ppm

m/z tolerance (ppm)

10

F) Saturated Ions

Enable saturated ions collapse ☒ ⓘ

Correlation $\geq$

0.8

rt tolerance (min)

0.05

m/z tolerance type:

☒ Da ☐ ppm

Max m/z distance (Da)

0.95

Min anchor intensity

100000

Min fold difference (Anchor/Artifact)

10

☐ Bidirectional search

☐ Show isotope/simer groups

☐ Show adduct groups

☐ Show adduct statistics

☐ Show neutral loss hits

☐ Show fragment hits

☐ Show mispicking ions groups

☐ Show saturated ions groups

Apply

Clear

3

3. Enable & Set Up Neutral Losses Filtering

Need parameter help? Click the tooltips or read the [About](#) tab for details. **Pro-Tip:** Use if only MS1 is available.

4

4. Enable & Set Up In-Source Fragments Filtering

Need parameter help? Click the tooltips or read the [About](#) tab for details. **Pro-Tip:** Use with “**Require MS2 fragment match**” option if MS2 data as is available. For this upload MGF file corresponding to the peak table.

5

5. Enable & Set Up Mispicked Ions Filtering

Need parameter help? Click the tooltips or read the [About](#) tab for details.

6

6. Enable & Set Up Saturated Ions Filtering

Need parameter help? Click the tooltips or read the [About](#) tab for details.



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Tab “QC Filters”: Control Panel

The screenshot shows the QC Filters control panel with the following components:

- Sample Label:** A section for specifying the source of sample labels, including options like "Use labels from Blank Tab", "From sample names (by token)", "From metadata CSV", "From custom CSV (one column, no header)", and "Manual editable table".
- QC value filters:** A section for enabling filters (Zeros, Mean, RSD, Min) and setting evaluation modes (Groupwise, EVERY, POOLED).
- Group(s) to evaluate:** A text input field for specifying groups.
- Zeros metric:** A section for setting the Zeros threshold (count or %) and the Mean threshold.
- RSD threshold:** A section for setting the RSD threshold.
- Min threshold:** A section for setting the Min threshold.
- Plot values distribution:** A checkbox to show the QC plot panel.
- Apply and Clear buttons:** Buttons to apply the filters or clear them.

Numbered callouts 1-7 point to the following features:

1. Sample Labels — how to specify groups
2. Show resulted Labels in Display Area
3. Set Up QC Values Filters
4. Set Up Zero Values Filters
5. Set Up Mean/RSD/Min Values Filters
6. Plot distribution of selected values
7. Apply or Discard Filtering Step

The output table shows the results of the filtering process, including the number of features before and after each filter step.

Metric	Value
Features before	2597
Removed by Zeros	5
Removed by Mean	1453
Removed by RSD	110
Removed by Min	595
Total removed	1485
Features after	1112

1. Sample Labels — how to specify groups

Choose a label source: from *Blank Filters* tab, sample names, metadata CSV, custom CSV (one column, no header), or by manual editing.

See details for extracting Labels on the Page **Tab “Blank Filters”: Control Panel**.

2. Show resulted Labels in Display Area

3. Set Up QC Values Filters

Select desired values for filtering: Zeros, Mean, RSD, or Min.

Choose between filtering modes:

- Groupwise (ANY selected group passes)
- Groupwise (EVERY selected group passes)
- POOLED across selected groups

Specify groups used for evaluation.

4. Set Up Zero Values Filters

Specify Zero metric (counts or %) and threshold.

5. Set Up Mean/RSD/Min Values Filters

Specify thresholds for corresponding value.

6. Plot distribution of selected values

7. Apply or Discard Filtering Step



Tab “QC Filters”: Display Area

Upload DataBlank FiltersMS FiltersQC FiltersPeak FiltersFinal SummaryAbout

Sample Label

Label source:

- Use labels from Blank Tab
- From sample names (by token)
- From metadata CSV
- From custom CSV (one column, no header)
- Manual editable table

☒ Show labels table

QC value filters

Enable filters:

ZerosMeanRSDMin

Evaluation mode:

- Groupwise (ANY selected group passes)
- Groupwise (EVERY selected group passes)
- POOLED across selected groups

Group(s) to evaluate:

Empty = all groups

Zeros metric

- Count
- Percent

Zeros $\leq$ threshold (count or %)

0

Mean $\geq$ threshold

5000

RSD $\leq$ threshold

30

Min $\geq$ threshold

1000

Plot values distribution

☒ Show QC plot panel

Apply

Clear

Input table

Show 5 entries

Search:

row ID	mz	rt	row ion mobility unit	row CCS	correlation group ID	annotat network number
1	108	148.0277472	0.3691534		31	0
2	181	169.0462455	0.3699988		31	
4	231	216.951			31	
5	43	234.961351	0.3725722		31	14
6	145	244.9475245	0.37269926		31	

Showing 1 to 5 of 2597 entries

Previous12345...520Next

Check matching Sample Names and Group Labels:

Show 6 entries

Search:

Sample	Label
Orbi_Blank_A.mzML Peak area	Blank
Orbi_Blank_B.mzML Peak area	Blank
Orbi_Blank_C.mzML Peak area	Blank
Orbi_Sample_A.mzML Peak area	Sample
Orbi_Sample_B.mzML Peak area	Sample
Orbi_Sample_C.mzML Peak area	Sample

Showing 1 to 6 of 9 entries

Previous12Next

Check Values Distribution:

ZEROS [Zero values (count)] - Groupwise (ANY) | Filtered: 5 of 2597 (0.2%)

MEAN [Mean intensity (log10)] - Groupwise (ANY) | Filtered: 1453 of 2597 (55.9%)

MIN [Minimum intensity (log10)] - Groupwise (ANY) | Filtered: 595 of 2597 (22.9%)

RSD [RSD (%)] - Groupwise (ANY) | Filtered: 366 of 2597 (14.1%)

Output table — after QC Filters

Show 5 entries

Search:

row ID	mz	rt	row ion mobility unit	row CCS	correlation group ID	annotat network number
1	108	148.0277472	0.3691534		31	0
2	181	169.0462455	0.3699988		31	
4	231	216.951			31	
5	43	234.961351	0.3725722		31	14
6	145	244.9475245	0.37269926		31	

Showing 1 to 5 of 1112 entries

Previous12345...223Next

Filtering results — Step 3 (QC Filters)

Metric	Value
Features before	2597
Removed by Zeros	5
Removed by Mean	1453
Removed by RSD	
Removed by Min	
Total removed	5
Features after	1112

Note: a feature may be counted in multiple filter categories.



Tab “Peak Filters”: Control Panel

The screenshot shows the 'Peak Filters' control panel with five numbered callouts:

- 1. Set Up Peak Values Filters**: Points to the 'Enable filters' section where 'm/z', 'rt', 'RMD', and 'AMD' can be selected.
- 2. Set Up m/z /rt/AMD/RMD Values Filters**: Points to the input fields for 'm/z $\geq$ ' (set to 200) and 'rt $\geq$ (min)' (set to 1).
- 3. Set Up Peaks to Remove/Keep**: Points to the 'Action for matched target peaks' section, where 'Remove matched peaks' is selected.
- 4. Plot distribution of selected values**: Points to the 'Plot values distribution' checkbox, which is checked.
- 5. Apply or Discard Filtering Step**: Points to the 'Apply' button.

The interface also includes an 'Input table' showing 5 entries, a 'Matched Peaks' table showing 2 entries, and a scatter plot of 'Filtered m/z: 252' vs 'RT (min)'. The 'Output table – after Peak Filters' shows 5 entries. A 'Filtering results – Step 4 (Peak Filters)' summary table is at the bottom.

Metric	Value
Features before	2597
Removed by m/z	252
Removed by rt	555
Removed by RMD	0
Removed by AMD	0
Matched target peaks	2
Removed by target $m/z + rt$	2
Total removed	656
Features after	1941

Note: a feature may be counted in multiple filter categories.

1. Set Up Peak Values Filters

Select desired values for filtering:

m/z , rt , mass defect (RMD, or AMD), or a list of target peaks to remove.

2. Set Up m/z /rt/AMD/RMD Values Filters

Specify threshold ranges for corresponding value.

3. Set Up Peaks to Remove/Keep

Specify action as keep or remove; provide target m/z and RT pairs. Enter comma-separated values manually, one pair per line, or upload a CSV file. Specify m/z and RT tolerances for matching.

4. Plot distribution of selected values

5. Apply or Discard Filtering Step



Tab “Peak Filters”: Display Area

MC

ADAMC

Upload DataBlank FiltersMS FiltersQC FiltersPeak FiltersFinal SummaryAbout

Peak Filters

Enable filters: 1

m/zrtRMDAMD

Target peak list

☒ m/z ≥

☐ m/z ≤

200

☒ rt ≥ (min)

☐ rt ≤ (min)

1

Action for matched target peaks:

☒ Remove matched peaks

☐ Keep only matched peaks

m/z tolerance type:

☒ Da☐ ppm

m/z tolerance (Da)

0.005

rt tolerance (min)

0.05

Type target peaks:

506.528, 11.6

970.611, 9.7

Or upload target list (.csv)

Browse...

No file selected

CSV should contain m/z and rt columns. If column names are not detected, the first two columns are treated as m/z and rt.

Plot values distribution

☒ Show Peak plot panel

Apply

Clear

Input table

Show 5 entries

Search:

row ID	m/z	rt	row ID	row ion mobility	row CCS	correlation group ID	annotation number
1	108	148.0277472	0.3691534			31	0
2	181	169.0462455	0.3699988			31	
4	231	216.951263				31	
5	43	234.96135				31	14
6	145	244.9475245	0.37269926			31	

Showing 1 to 5 of 2,597 entries

Previous12345...520Next

Matched Peaks

Show 5 entries

Search:

target_id	target_mz	target_rt	m/z	rt	delta_mz	delta_mz_ppm	
typed_1	506.5280	11.600	24	506.5288	11.598	0.0008	1.5
ty							0.4
1.4							1.2

Showing 1 to 5 of 2,597 entries

Previous12345...520Next

Check values map:

m/z vs RT

1500

1000

500

0

3

6

9

12

RT (min)

Filtered m/z: 252
Filtered rt: 555
Total Filtered: 654

Output table – after Peak Filters

Show 5 entries

Search:

row ID	m/z	rt	row ID	row ion mobility	row CCS	correlation group ID	annotation number
818	2815	336.074296	1.0103563			58	489
819	2820	314.0921816	1.0123757			58	489
820	2814	284.134450					490
821	2819	289.09139					490
822	2824	352.0481904					489

Showing 1 to 5 of 1,941 entries

Previous12345...389Next

Filtering results – Step 4 (Peak Filters)

Metric	Value
Features before	2597
Removed by m/z	252
Removed by rt	555
Removed by RMD	0
Removed by AMD	0
Matched target peaks	2
Removed by target m/z + rt	2
Total removed	656
Features after	1941



Tab “Final Summary”: Control Panel



Upload Data

Blank Filters

MS Filters

QC Filters

Peak Filters

Final Summary

About

Compile final output

Compile output ?

Final peak table (.csv)

MGF Filtering

Filter MGF ☒ ?

Upload MGF file

Browse...

orbi_iimn_gnps.mgf

Upload complete

Select ID column in Table:

row ID

MGF Feature ID field:

acquisitionNum

Filter MGF

Clear MGF

Success! Matched 321 spectra from your filtered table.

Download Filtered MGF

Filtering summary

- Dataset: Example
- Step 1 Blank Filters: APPLIED
- Step 2 MS Filters: APPLIED

1. Compile final filtered table

Clicking this button enables the table download and MGF filtering.

2. Download final table

3. Enable MGF filtering & upload .mgf file

4. Match peak ID between table and .mgf file

The feature identifier in the final peak table must correspond to the chosen spectrum identifier field inside the uploaded MGF. We recommend using the default columns unless you have specific requirements.

5. Filter or clear uploaded .mgf file

6. Download filtered .mgf file

Showing 1 to 5 of 1,083 entries

Previous

1

2

3

4

5

...

217

Next

819		489
821		490
822		489
829		495
830		495

18



Tab “Final Summary”: Display Area

[Upload Data](#)[Blank Filters](#)[MS Filters](#)[QC Filters](#)[Peak Filters](#)[Final Summary](#)[About](#)

Compile final output

[Compile output](#)[Final peak table \(.csv\)](#)

MGF Filtering

Filter MGF ☒

Upload MGF file

[Browse...](#)

orbi_iimn_gnps.mgf

Upload complete

Select ID column in Table:

row ID

MGF Feature ID field:

acquisitionNum

[Filter MGF](#)[Clear MGF](#)

Success! Matched 321 spectra from your filtered table.

[Download Filtered MGF](#)

Filtering summary

- Dataset: Example
- Step 1 Blank Filters: APPLIED
- Step 2 MS Filters: APPLIED
- Step 3 QC Filters: not applied
- Step 4 Peak Filters: APPLIED
- Raw features: 6167
- Filtered features: 5084 (82.44%)
- Final features: 1083

Final table

Show 5 entries

Search:

	row ID	row m/z	row retention time	row ion mobility	row ion mobility unit	row CCS	correlation group ID	annotation network number
819	2820	314.0921816	1.0127777				58	489
821	2819	289.091359	1.016				58	490
822	2824	352.0481904	1.0176588				58	489
829	2893	275.1104506	1.0585195				60	495
830	2862	253.1291676	1.0596385				60	495

Showing 1 to 5 of 1,083 entries

[Previous](#)[1](#)[2](#)[3](#)[4](#)[5](#)[...](#)[217](#)[Next](#)

3 Matches in filtered .mgf



Tab “About” Overview

How to:

Select a section:

- Quick Start
- Upload Data
- Blank Filters
- MS Filters
- QC Filters
- Peak Filters
- Final Summary
- Troubleshooting
- Project Details

Troubleshooting

- **No sample columns:** your sample keywords don't match sample column names.
- **Parsing error:** check selected Data table type and compatibility with examples of accessible Peak Table data formats in the [GitHub](#) repository.
- **Disconnected from the server:** could be caused by memory limit, try to compile and run it locally. Instructions in the [GitHub](#) repository.

Navigate through the sections to get more information